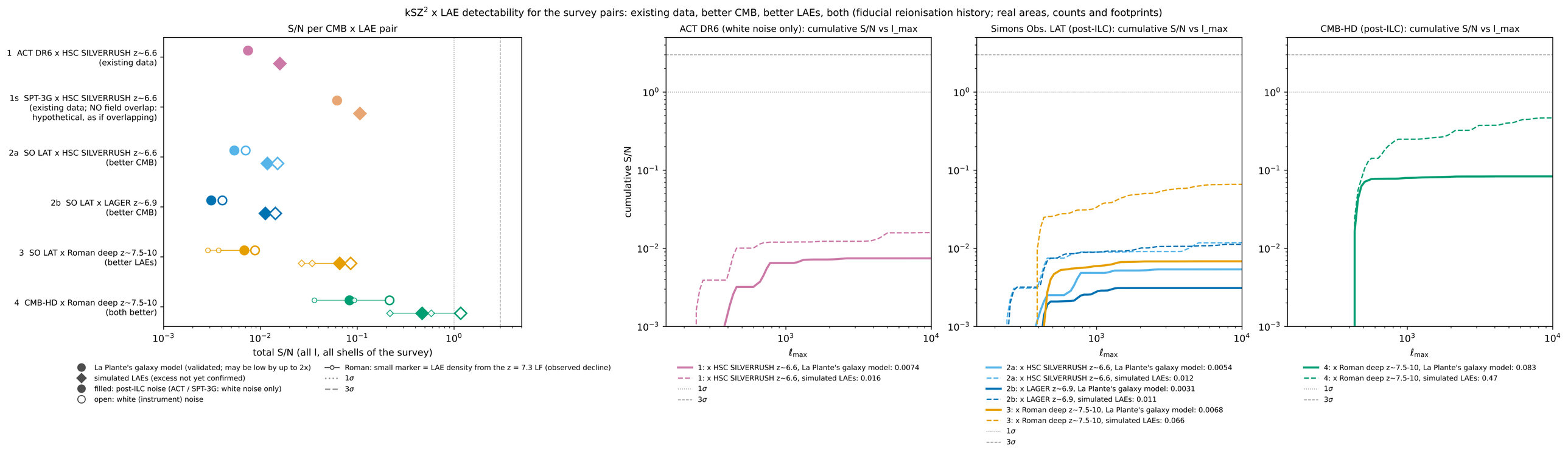
kSZ² x LAE detectability for the survey pairs: existing data, better CMB, better LAEs, both (fiducial reionisation history; real areas, counts and footprints)
S/N per CMB x LAE pair
1 ACT DR6 x HSC SILVERRUSH z~6.6 (existing data)
1s SPT-3G x HSC SILVERRUSH z~6.6 (existing data; NO field overlap: hypothetical, as if overlapping)
2a SO LAT x HSC SILVERRUSH z~6.6 (better CMB)
2b SO LAT x LAGER z~6.9 (better CMB)
3 SO LAT x Roman deep z~7.5-10 (better LAEs)
4 CMB-HD x Roman deep z~7.5-10 (both better)
total S/N (all l, all shells of the survey)
La Plante's galaxy model (validated; may be low by up to 2x)
simulated LAEs (excess not yet confirmed)
filled: post-ILC noise (ACT / SPT-3G: white noise only)
open: white (instrument) noise
Roman: small marker = LAE density from the z = 7.3 LF (observed decline)
1σ
3σ
ACT DR6 (white noise only): cumulative S/N vs l_max
Simons Obs. LAT (post-ILC): cumulative S/N vs l_max
CMB-HD (post-ILC): cumulative S/N vs l_max
cumulative S/N
ℓmax
1: x HSC SILVERRUSH z~6.6, La Plante's galaxy model: 0.0074
1: x HSC SILVERRUSH z~6.6, simulated LAEs: 0.016
2a: x HSC SILVERRUSH z~6.6, La Plante's galaxy model: 0.0054
2a: x HSC SILVERRUSH z~6.6, simulated LAEs: 0.012
2b: x LAGER z~6.9, La Plante's galaxy model: 0.0031
2b: x LAGER z~6.9, simulated LAEs: 0.011
3: x Roman deep z~7.5-10, La Plante's galaxy model: 0.0068
3: x Roman deep z~7.5-10, simulated LAEs: 0.066
4: x Roman deep z~7.5-10, La Plante's galaxy model: 0.083
4: x Roman deep z~7.5-10, simulated LAEs: 0.47